

Electromagnetic Theory

Coulomb's Law

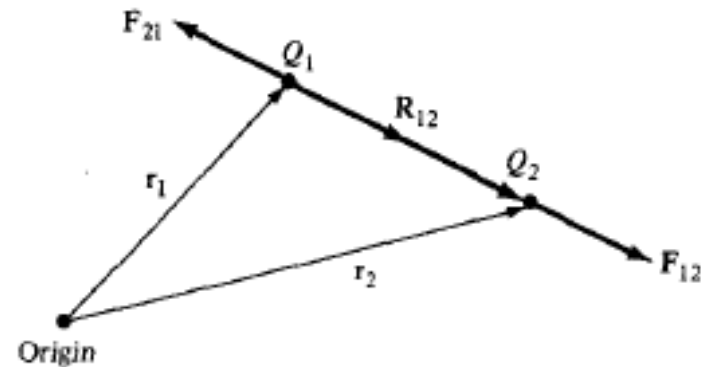
It states that the force F between two point charges Q_1 and Q_2 is

$$F = \frac{kQ_1Q_2}{R^2}$$

In Vector form

$$\mathbf{F}_{12} = \frac{Q_1Q_2}{4\pi\epsilon_0 R^3} \mathbf{R}_{12}$$

Or
$$\mathbf{F}_{12} = \frac{Q_1Q_2 (\mathbf{r}_2 - \mathbf{r}_1)}{4\pi\epsilon_0 |\mathbf{r}_2 - \mathbf{r}_1|^3}$$



If we have more than two point charges

$$\mathbf{F} = \frac{Q}{4\pi\epsilon_0} \sum_{k=1}^N \frac{Q_k(\mathbf{r} - \mathbf{r}_k)}{|\mathbf{r} - \mathbf{r}_k|^3}$$

Electric Field Intensity

Electric Field Intensity is the force per unit charge when placed in the electric field

$$E = \frac{F}{Q}$$

In Vector form

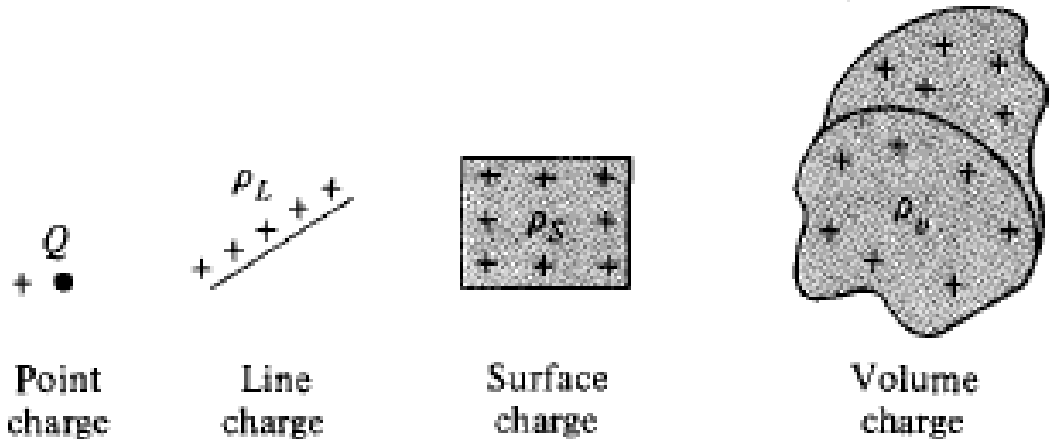
$$\mathbf{E} = \frac{Q(\mathbf{r} - \mathbf{r}')}{4\pi\epsilon_0|\mathbf{r} - \mathbf{r}'|^3}$$

If we have more than two point charges

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \sum_{k=1}^N \frac{Q_k(\mathbf{r} - \mathbf{r}_k)}{|\mathbf{r} - \mathbf{r}_k|^3}$$

Electric Field due to Continuous Charge Distribution

If there is a continuous charge distribution say along a line, on a surface, or in a volume



The charge element dQ and the total charge Q due to these charge distributions can be obtained by

$$dQ = \rho_L dl \rightarrow Q = \int_L \rho_L dl \quad (\text{line charge})$$

$$dQ = \rho_S dS \rightarrow Q = \int_S \rho_S dS \quad (\text{surface charge})$$

$$dQ = \rho_v dv \rightarrow Q = \int_v \rho_v dv \quad (\text{volume charge})$$

The electric field intensity due to each charge distribution ρ_L , ρ_S and ρ_V may be given by the summation of the field contributed by the numerous point charges making up the charge distribution.

$$\mathbf{E} = \int \frac{\rho_L dl}{4\pi\epsilon_0 R^2} \mathbf{a}_R \quad (\text{line charge})$$

$$\mathbf{E} = \int \frac{\rho_S dS}{4\pi\epsilon_0 R^2} \mathbf{a}_R \quad (\text{surface charge})$$

$$\mathbf{E} = \int \frac{\rho_v dv}{4\pi\epsilon_0 R^2} \mathbf{a}_R \quad (\text{volume charge})$$

Electric Field Intensity due to Line Charge

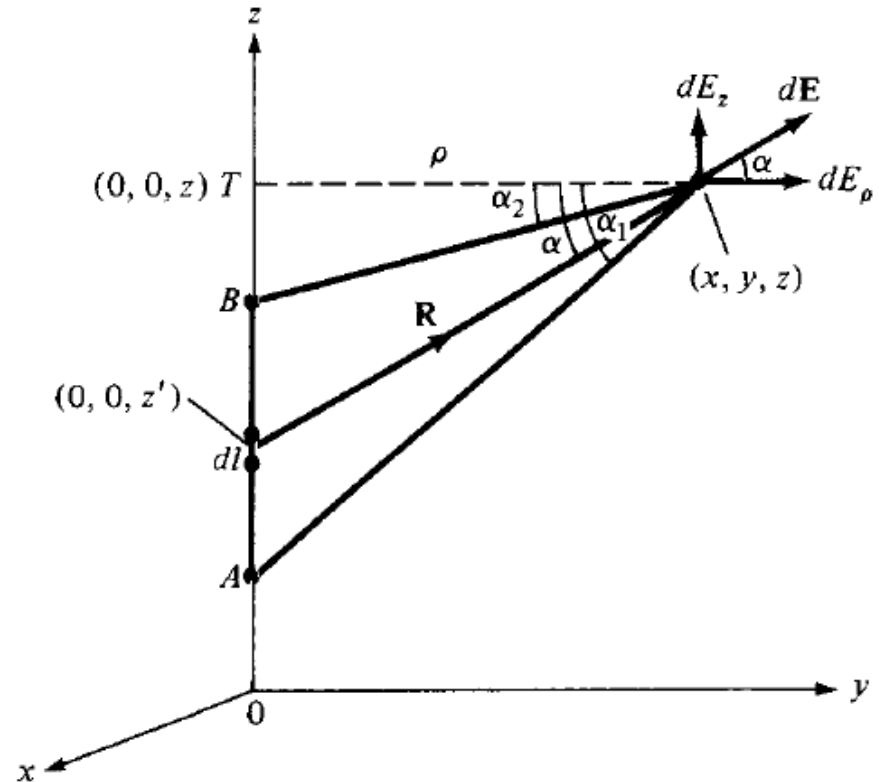
Consider a line charge with uniform charge density ρ_L extending from A to B along the z-axis.

The charge element dQ associated with element $dl = dz$ of the line is

$$dQ = \rho_L dl = \rho_L dz$$

The total charge Q is

$$Q = \int_{z_A}^{z_B} \rho_L dz$$



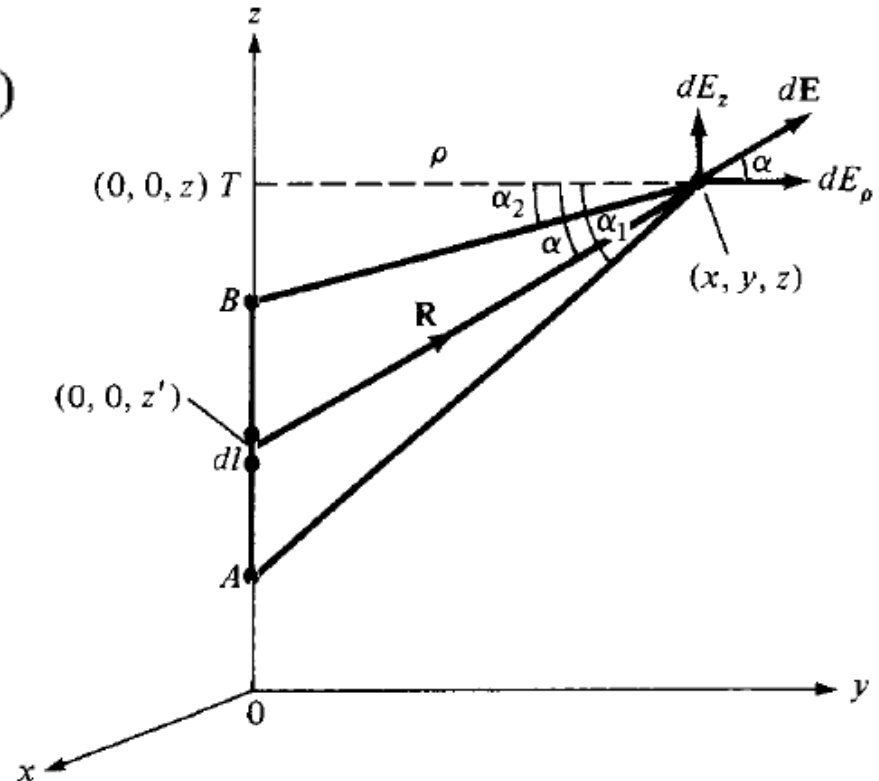
Electric Field Intensity due to Line Charge

The electric field intensity E at an arbitrary point $P (x, y, z)$ can be given by

$$\mathbf{E} = \int \frac{\rho_L dl}{4\pi\epsilon_0 R^2} \mathbf{a}_R \quad (\text{line charge})$$

The field point is generally denoted by (x, y, z) and the source point as (x', y', z') . So

$$dl = dz'$$



Electric Field Intensity due to Line Charge

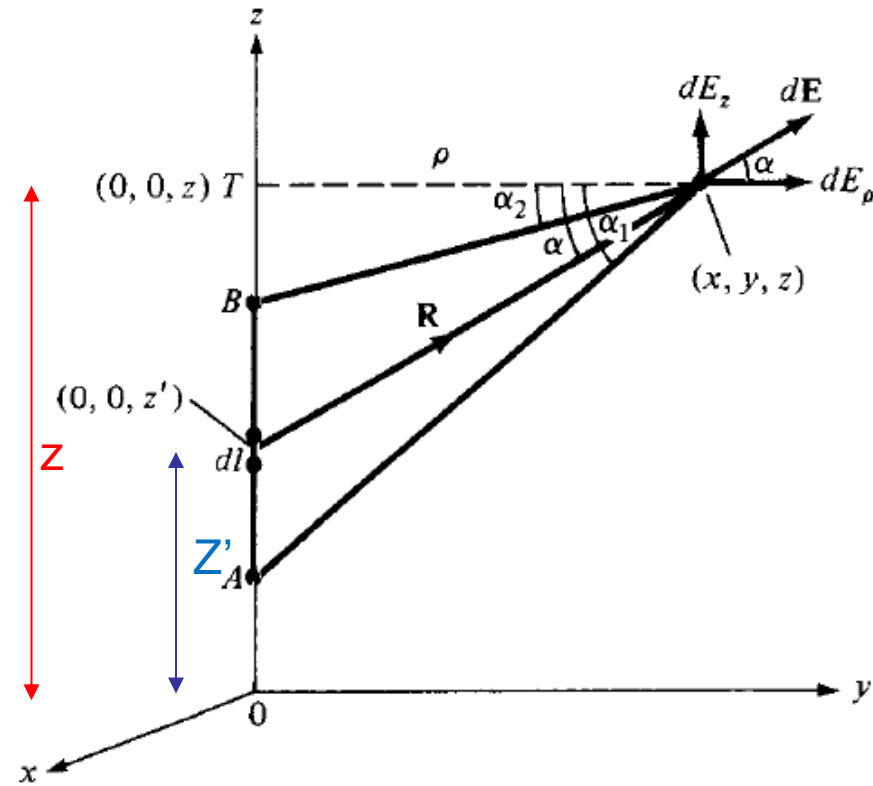
$$\mathbf{R} = \rho \mathbf{a}_\rho + (z - z') \mathbf{a}_z$$

$$R^2 = |\mathbf{R}|^2 = \rho^2 + (z - z')^2$$

$$\frac{\mathbf{a}_R}{R^2} = \frac{\mathbf{R}}{|\mathbf{R}|^3} = \frac{\rho \mathbf{a}_\rho + (z - z') \mathbf{a}_z}{[\rho^2 + (z - z')^2]^{3/2}}$$

Hence equation becomes

$$\mathbf{E} = \frac{\rho_L}{4\pi\epsilon_0} \int \frac{\rho \mathbf{a}_\rho + (z - z') \mathbf{a}_z}{[\rho^2 + (z - z')^2]^{3/2}} dz'$$



Electric Field Intensity due to Line Charge

To evaluate last equation we define $\alpha, \alpha_1, \alpha_2$.

$$R = [\rho^2 + (z - z')^2]^{1/2} = \rho \sec \alpha$$

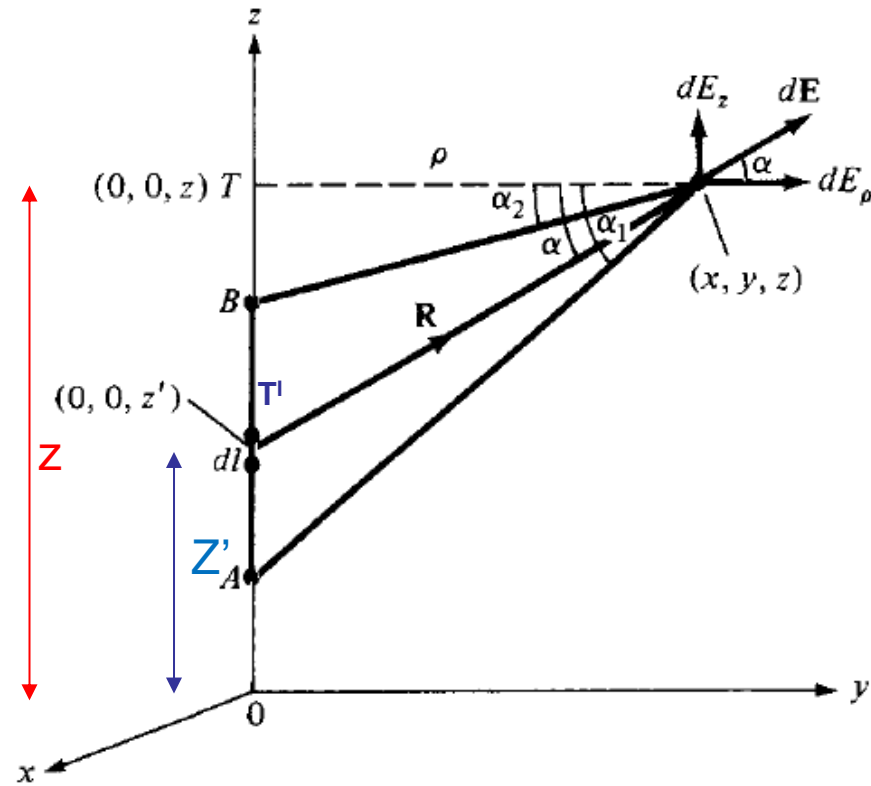
$$Z' = OT - TT'$$

$$Z' = Z - TT' \Rightarrow Z - Z' = TT'$$

$$z' = OT - \rho \tan \alpha$$

$$dz' = -\rho \sec^2 \alpha d\alpha$$

Hence Electric Field Intensity equation becomes



$$\mathbf{E} = \frac{\rho_L}{4\pi\epsilon_0} \int_{\alpha_1}^{\alpha_2} \frac{[\rho a_\rho + (\rho \tan \alpha) a_z]}{\rho^3 \sec^3 \alpha} \rho \sec^2 \alpha d\alpha$$

$$E = \frac{\rho_L}{4\pi\epsilon_0} \int \frac{\rho a_p + (z-z')a_z}{[R^2 + (z-z')^2]^{3/2}} dz'$$

$$= \frac{\rho_L}{4\pi\epsilon_0} \int \frac{\rho a_p + (z-z')a_z}{R^3}$$

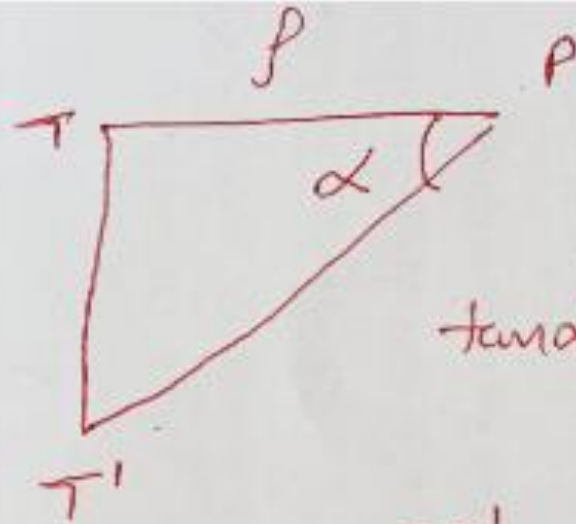
$$z-z' = \rho \tan \alpha$$

$$dz' = -\rho \sec^2 \alpha d\alpha$$

$$R = \rho \sec \alpha$$

$$= \frac{\rho_L}{4\pi\epsilon_0} \int_{\alpha_1}^{\alpha_2} \frac{[\rho a_p + (\rho \tan \alpha) a_z]}{\rho^3 \sec^3 \alpha} \rho \sec^2 \alpha d\alpha$$

$$= \frac{\rho_L}{4\pi\epsilon_0} \int_{\alpha_1}^{\alpha_2} \frac{a_p + \tan \alpha a_z}{\sec \alpha} d\alpha$$



$$\tan \alpha = \frac{TT'}{TP}$$

$$TT' = f \tan \alpha$$

$$\cos \alpha = \frac{f}{R} \Rightarrow f = \underline{R \cos \alpha}$$

$$\boxed{R = f \sec \alpha}$$

Electric Field Intensity due to Line Charge

$$= -\frac{\rho_L}{4\pi\epsilon_0\rho} \int_{\alpha_1}^{\alpha_2} [\cos \alpha \mathbf{a}_\rho + \sin \alpha \mathbf{a}_z] d\alpha$$

For finite line charge

$$\mathbf{E} = \frac{\rho_L}{4\pi\epsilon_0\rho} [-(\sin \alpha_2 - \sin \alpha_1)\mathbf{a}_\rho + (\cos \alpha_2 - \cos \alpha_1)\mathbf{a}_z]$$

For an infinite line charge, point B is at $(0, 0, \infty)$ and A at $(0, 0, -\infty)$ then

$$\alpha_1 = \pi/2, \alpha_2 = -\pi/2$$

and

$$\mathbf{E} = \frac{\rho_L}{2\pi\epsilon_0\rho} \mathbf{a}_\rho$$

Electric Flux Density

The electric field intensity depends on the medium in which the charges are placed.

Suppose a vector field $\mathbf{D}$ independent of the medium is defined by

$$\mathbf{D} = \epsilon_0 \mathbf{E}$$

The electric flux ψ in terms of $\mathbf{D}$ can be defined as

$$\Psi = \int \mathbf{D} \cdot d\mathbf{S}$$

The vector field $\mathbf{D}$ is called the electric flux density and is measured in coulombs per square meter.

GAUSS LAW

It states that the total electric flux ψ through any closed surface is equal to the total charge enclosed by that surface.

$$\psi = Q_{enc}$$

$$\Psi = \oint d\Psi = \oint_S \mathbf{D} \cdot d\mathbf{S}$$

$$\text{Total charge enclosed } Q = \int \rho_v dv$$

$$Q = \oint_S \mathbf{D} \cdot d\mathbf{S} = \int_v \rho_v dv \quad (i)$$

Using Divergence Theorem

$$\oint_S \mathbf{D} \cdot d\mathbf{S} = \int_v \nabla \cdot \mathbf{D} dv \quad (\text{ii})$$

Comparing the two volume integrals in (i) and (ii)

$$\rho_v = \nabla \cdot \mathbf{D}$$

This is the first Maxwell's equation.

It states that the volume charge density is the same as the divergence of the electric flux density.

Applications of Gauss Law (Point charge)

Let a point charge is located at Origin.

To determine $\mathbf{D}$ at a point P, consider a spherical surface centered at Origin.

$\mathbf{D}$ is everywhere normal to the Gaussian surface.

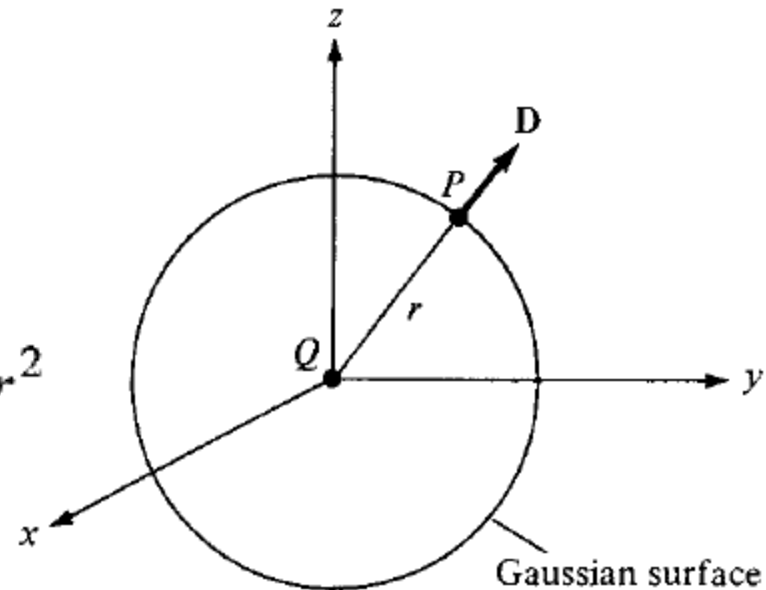
$$\mathbf{D} = D_r \mathbf{a}_r$$

$$Q = \oint \mathbf{D} \cdot d\mathbf{S} = D_r \oint dS$$

$$\oint dS = \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} r^2 \sin \theta d\theta d\phi = 4\pi r^2$$

$$Q = D_r 4\pi r^2$$

$$\text{Hence } \mathbf{D} = \frac{Q}{4\pi r^2} \mathbf{a}_r$$



Applications of Gauss Law (Infinite Line charge)

Consider the infinite line of uniform charge ρ_L C/m lies along the z-axis.

To determine $\mathbf{D}$ at a point P, consider a cylindrical surface centered at Origin.

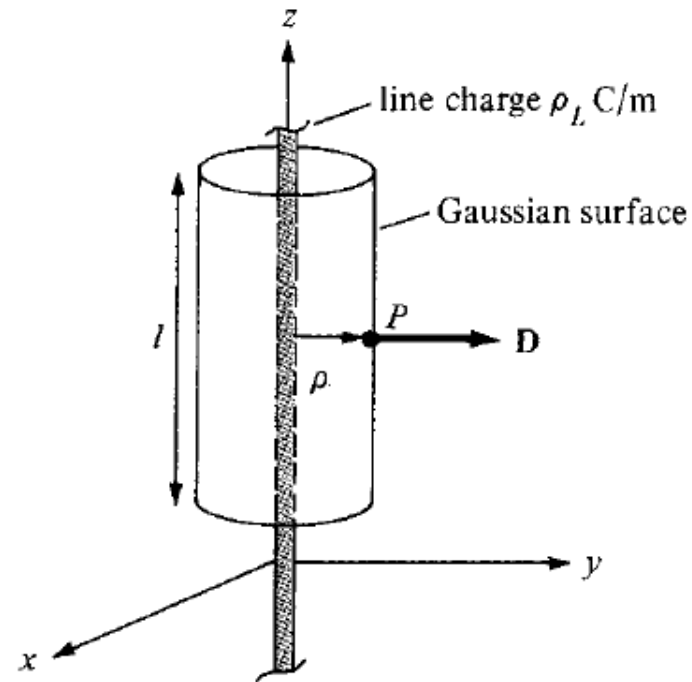
$\mathbf{D}$ is constant and normal to the cylindrical Gaussian surface.

$$\mathbf{D} = D_\rho \mathbf{a}_\rho$$

Applying Gauss law to an arbitrary length l

$$\rho_L \ell = Q = \oint \mathbf{D} \cdot d\mathbf{S} = D_\rho \oint dS$$

$$\text{Here } \oint dS = 2\pi\rho\ell$$

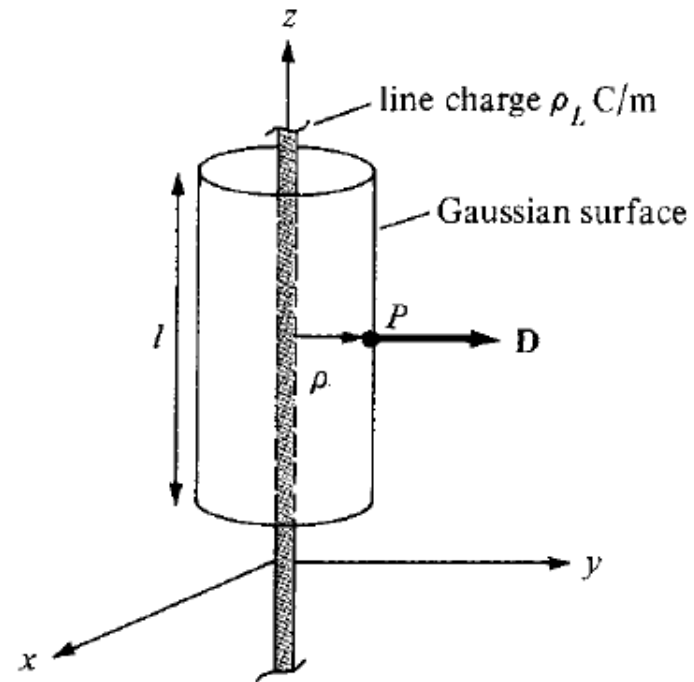


Applications of Gauss Law (Infinite Line charge)

$$\rho_L \ell = D_\rho 2\pi\rho\ell$$

$\int \mathbf{D} \cdot d\mathbf{S}$ evaluated on the top and bottom surfaces of the cylinder is zero since $\mathbf{D}$ has no z-component. It means $\mathbf{D}$ is tangential to the surfaces.

Hence
$$\mathbf{D} = \frac{\rho_L}{2\pi\rho} \mathbf{a}_\rho$$



Applications of Gauss Law (Infinite Sheet of charge)

Consider the infinite sheet of uniform charge ρ_S C/m² lying on the $z = 0$ plane.

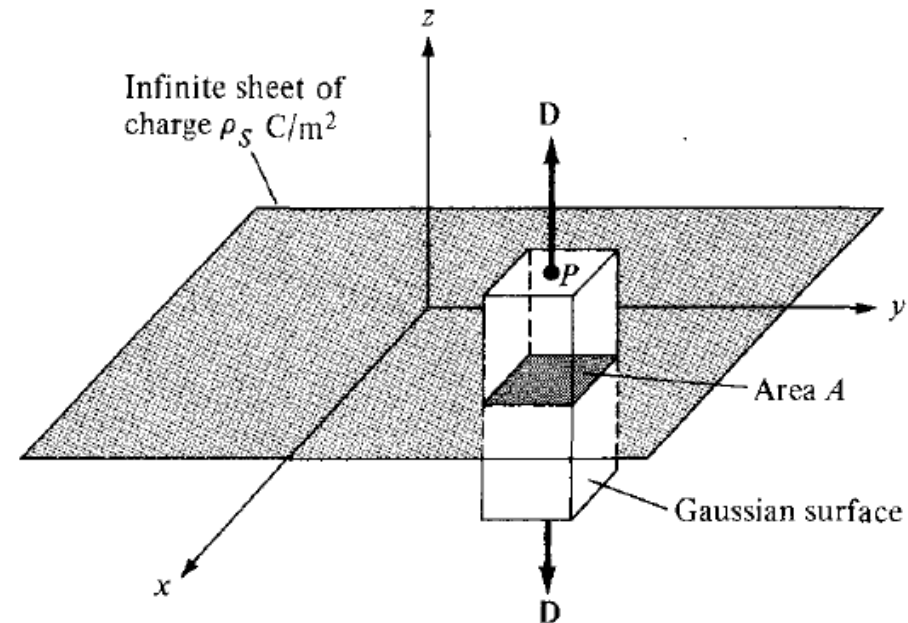
To determine $\mathbf{D}$ at point P, let's choose a rectangular box that is cut symmetrically by the sheet of charge and has two of its faces parallel to the sheet.

$\mathbf{D}$ is normal to the sheet

$$\mathbf{D} = D_z \mathbf{a}_z$$

Applying Gauss Law

$$\rho_S \int dS = Q = \oint \mathbf{D} \cdot d\mathbf{S} = D_z \left[\int_{\text{top}} dS + \int_{\text{bottom}} dS \right]$$



Applications of Gauss Law (Infinite Sheet of charge)

$\int \mathbf{D} \cdot d\mathbf{S}$ evaluated on the sides of the box is zero as $\mathbf{D}$ has no components along $\mathbf{a}_x$ and $\mathbf{a}_y$

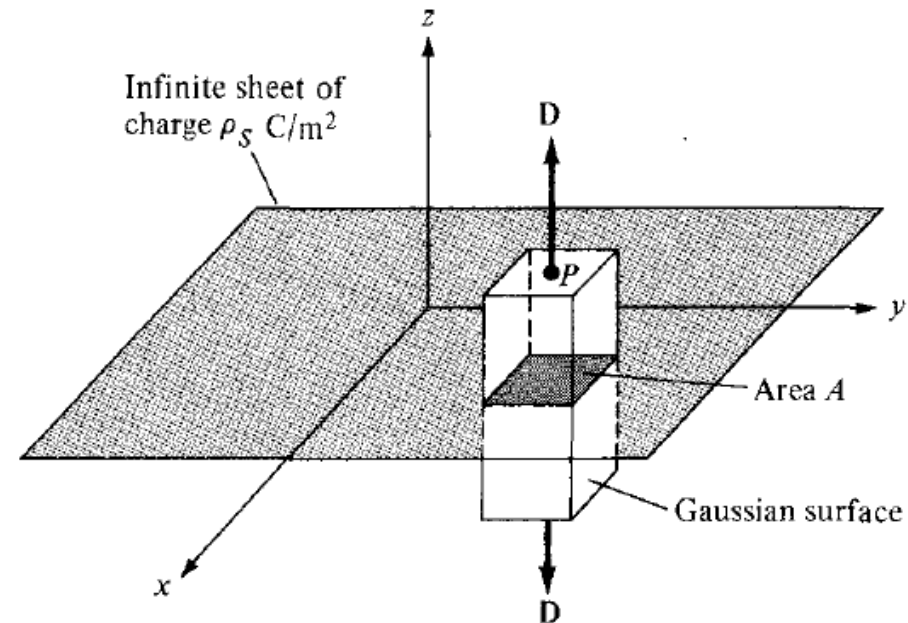
If the top and bottom of the box, each has area A then

$$\rho_s A = D_z (A + A)$$

Hence $\mathbf{D} = \frac{\rho_s}{2} \mathbf{a}_z$

or

$$\mathbf{E} = \frac{\mathbf{D}}{\epsilon_0} = \frac{\rho_s}{2\epsilon_0} \mathbf{a}_z$$



Electric Potential

Electric Field intensity, E due to a charge distribution can be obtained from Coulomb's Law.

or using Gauss Law when the charge distribution is symmetric.

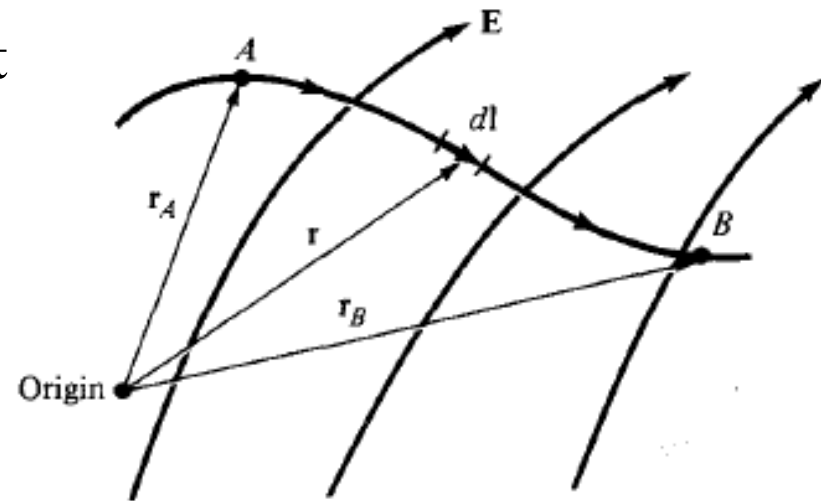
We can obtain E without involving vectors by using the electric scalar potential V .

From Coulomb's Law the force on point charge Q is

$$\vec{F} = Q\vec{E}$$

The work done in displacing the charge by length dl is

$$dW = \vec{F} \cdot d\vec{l} = Q\vec{E} \cdot d\vec{l}$$



The positive sign indicates that the work is being done by an external agent.

The total work done or the potential energy required in moving the point charge Q from A to B is

$$W = Q \int_A^B \vec{E} \cdot d\vec{l}$$

Dividing the above equation by Q gives the potential energy per unit charge.

$$\frac{W}{Q} = \int_A^B \vec{E} \cdot d\vec{l} = V_{AB}$$

V_{AB} is known as the potential difference between points A and B.

1. If V_{AB} is negative, there is loss in potential energy in moving Q from A to B (work is being done by the field), if V_{AB} is positive, there is a gain in potential energy in the movement (an external agent does the work).
2. It is independent of the path taken. It is measured in Joules per Coulomb referred as Volt.

The potential at any point due to a point charge Q located at the origin is

$$V = \frac{Q}{4\pi\epsilon_0 r}$$

The potential at any point is the potential difference between that point and a chosen point at which the potential is zero.

Assuming zero potential at infinity, the potential at a distance r from the point charge is the work done per unit charge by an external agent in transferring a test charge from infinity to that point.

$$V = -\int_{\infty}^r \vec{E} \cdot d\vec{l}$$

If the point charge Q is not at origin but at a point whose position vector is $\vec{r}'$, the potential $V(\vec{r})$ at $\vec{r}$ becomes

$$V(\vec{r}) = \frac{Q}{4\pi\epsilon_0 |\vec{r} - \vec{r}'|}$$

For n point charges $Q_1, Q_2, Q_3, \dots, Q_n$ located at points with position vectors $\vec{r}_1, \vec{r}_2, \vec{r}_3, \dots, \vec{r}_n$ the potential at $\vec{r}$ is

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \sum_{k=1}^n \frac{Q_k}{|\vec{r} - \vec{r}_k|}$$

If there is continuous charge distribution instead of point charges then the potential at $\vec{r}$ becomes

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int_L \frac{\rho_L(\mathbf{r}') dl'}{|\mathbf{r} - \mathbf{r}'|} \quad (\text{line charge})$$

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int_S \frac{\rho_S(\mathbf{r}') dS'}{|\mathbf{r} - \mathbf{r}'|} \quad (\text{surface charge})$$

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int_v \frac{\rho_v(\mathbf{r}') dv'}{|\mathbf{r} - \mathbf{r}'|} \quad (\text{volume charge})$$

The primed coordinates are used customarily to denote source point location and the unprimed coordinates refer to field point.

Relationship between E and V

The potential difference between points A and B is independent of the path taken

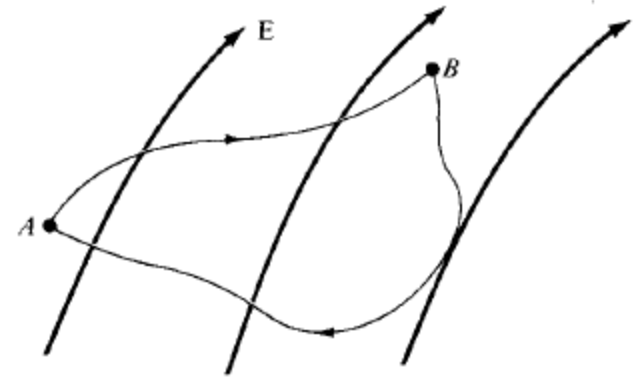
$$V_{AB} = -V_{BA}$$

$$V_{AB} = \int_A^B \vec{E} \cdot d\vec{l} \quad \text{and} \quad V_{BA} = -\int_B^A \vec{E} \cdot d\vec{l}$$

$$V_{AB} + V_{BA} = \oint \vec{E} \cdot d\vec{l} = 0$$

$$\oint \vec{E} \cdot d\vec{l} = 0 \quad (i)$$

It means that the line integral of $\vec{E}$ along a closed path must be zero.



Physically it means that no net work is done in moving a charge along a closed path in an electrostatic field.

Applying Stokes's theorem to equation (i)

$$\oint \vec{E} \cdot d\vec{l} = \int (\vec{\nabla} \times \vec{E}) \cdot d\vec{S} = 0$$

$$\vec{\nabla} \times \vec{E} = 0 \quad (\text{ii})$$

Equation (i) and (ii) are known as Maxwell's equation for static electric fields.

Equation (i) is in integral form while equation (ii) is in differential form, both depicting conservative nature of an electrostatic field.

Also

$$\vec{E} = -\nabla V$$

It means Electric Field Intensity is the gradient of V.

The negative sign shows that the direction of $\vec{E}$ is opposite to the direction in which V increases.

GRADIENT OF ELECTRIC POTENTIAL

Electric potential at any point A due to point charge q located at the origin O (Figure 4.25) such that $OA = \vec{r}$ is given by

$$V = \frac{q}{4\pi\epsilon_0 r}$$

Taking the gradient both sides we get

$$\vec{\nabla} V = \vec{\nabla} \left(\frac{q}{4\pi\epsilon_0 r} \right)$$

or
$$\vec{\nabla} V = \frac{q}{4\pi\epsilon_0} \vec{\nabla} \left(\frac{1}{r} \right)$$

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

$$r = \sqrt{x^2 + y^2 + z^2}$$

or
$$\frac{1}{r} = \frac{1}{\sqrt{x^2 + y^2 + z^2}} = (x^2 + y^2 + z^2)^{-1/2}$$

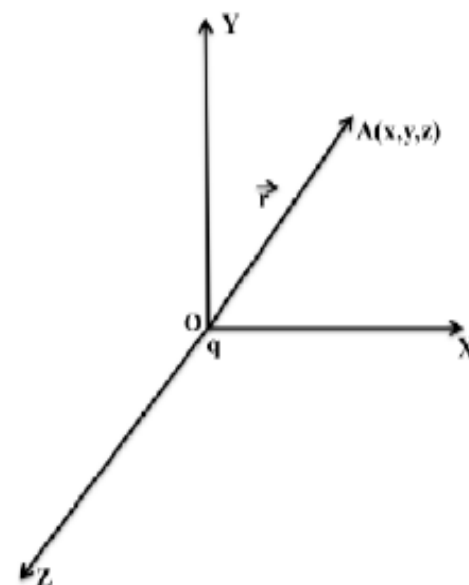


Figure 4.25

$$\begin{aligned}
 \therefore \vec{\nabla}\left(\frac{1}{r}\right) &= \left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}\right)(x^2 + y^2 + z^2)^{-1/2} \\
 &= -\frac{1}{2}(x^2 + y^2 + z^2)^{-3/2}(2x)\hat{i} - \frac{1}{2}(x^2 + y^2 + z^2)^{-3/2}(2y)\hat{j} - \frac{1}{2}(x^2 + y^2 + z^2)^{-3/2}(2z)\hat{k} \\
 &= -\frac{(x\hat{i} + y\hat{j} + z\hat{k})}{(x^2 + y^2 + z^2)^{3/2}} = -\frac{\vec{r}}{r^3} = -\frac{\hat{r}}{r^2}
 \end{aligned}$$

Equation (4.17.1) become

$$\vec{\nabla}V = \frac{q}{4\pi\epsilon_o} \vec{\nabla}\left(\frac{1}{r}\right) = -\frac{q}{4\pi\epsilon_o r^2} \hat{r}$$

$$\text{or } \vec{\nabla}V = -\frac{q}{4\pi\epsilon_o r^2} \hat{r} \quad (4.17.2)$$

Electric field due charge q at the observation point A is

$$\vec{E} = \frac{q}{4\pi\epsilon_o r^2} \hat{r} \quad (4.17.3)$$

By comparing Equation (4.17.2) and (4.17.3)

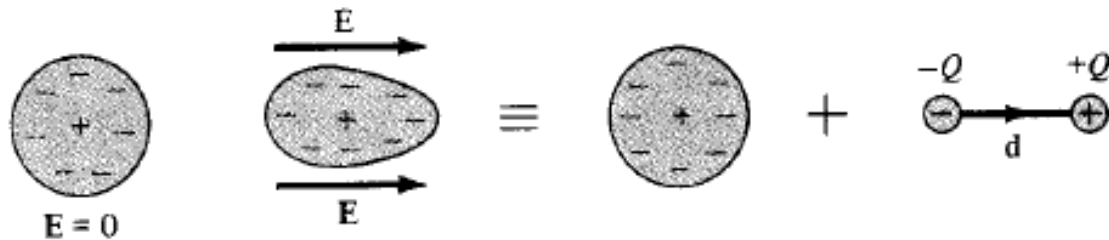
$$\begin{aligned}\vec{\nabla}V &= -\vec{E} \\ \Rightarrow \vec{E} &= -\vec{\nabla}V\end{aligned}\tag{4.17.4}$$

Equation (4.17.4) gives the relation between electric field and electric potential, which states that electric field at a point, is equal to the negative gradient of the electric potential which is true for point source as well as all type of charge distributions.

Polarization in Dielectrics

Consider an atom of the dielectric consisting of an electron cloud ($-Q$) and a positive nucleus ($+Q$).

When an electric field $\vec{E}$ is applied, the positive charge is displaced from its equilibrium position in the direction of $\vec{E}$ by $\vec{F}_+ = Q\vec{E}$ while the negative charge is displaced by $\vec{F}_- = Q\vec{E}$ in the opposite direction.



A dipole results from the displacement of charges and the dielectric is polarized. In polarized the electron cloud is distorted by the applied electric field.

This distorted charge distribution is equivalent to the original distribution plus the dipole whose moment is

$$\vec{p} = Q\vec{d}$$

where $\vec{d}$ is the distance vector between $-Q$ to $+Q$.

If there are N dipoles in a volume Δv of the dielectric, the total dipole moment due to the electric field

$$Q_1\mathbf{d}_1 + Q_2\mathbf{d}_2 + \cdots + Q_N\mathbf{d}_N = \sum_{k=1}^N Q_k\mathbf{d}_k$$

For the measurement of intensity of polarization, we define polarization $\vec{P}$ (coulomb per square meter) as dipole moment per unit volume

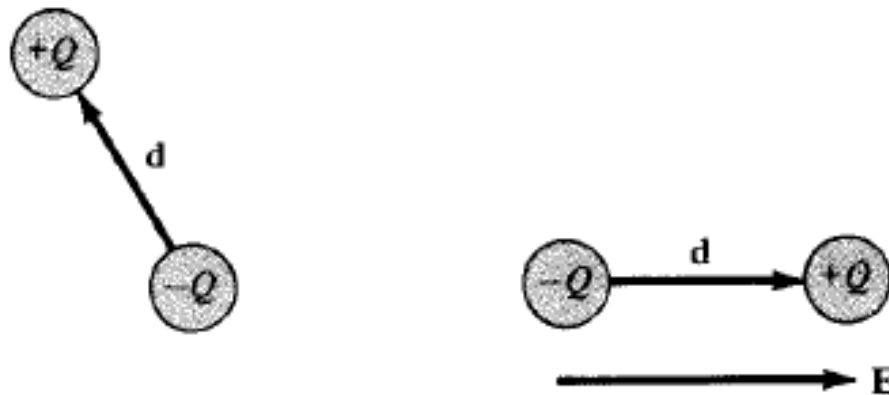
$$\mathbf{P} = \lim_{\Delta v \rightarrow 0} \frac{\sum_{k=1}^N Q_k\mathbf{d}_k}{\Delta v}$$

The major effect of the electric field on the dielectric is the creation of dipole moments that align themselves in the direction of electric field.

This type of dielectrics are said to be non-polar. eg: H_2 , N_2 , O_2

Other types of molecules that have in-built permanent dipole moments are called polar. eg: H_2O , HCl

When electric field is applied to a polar material then its permanent dipole experiences a torque that tends to align its dipole moment in the direction of the electric field.



Dielectric Constant and Strength

We know that

$$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P} \quad \text{and} \quad \mathbf{P} = \chi_e \epsilon_0 \mathbf{E}$$

Thus

$$\mathbf{D} = \epsilon_0 (1 + \chi_e) \mathbf{E} = \epsilon_0 \epsilon_r \mathbf{E}$$

or

$$\mathbf{D} = \epsilon \mathbf{E}$$

where

$$\epsilon = \epsilon_0 \epsilon_r$$

and

$$\epsilon_r = 1 + \chi_e = \frac{\epsilon}{\epsilon_0}$$

where ϵ is the permittivity of the dielectric, ϵ_0 is the permittivity of the free space and ϵ_r is the dielectric constant or relative permittivity.

No dielectric is ideal. When the electric field in a dielectric is sufficiently high then it begins to pull electrons completely out of the molecules, and the dielectric becomes conducting.

When a dielectric becomes conducting then it is called dielectric breakdown. It depends on the type of material, humidity, temperature and the amount of time for which the field is applied.

The minimum value of the electric field at which the dielectric breakdown occurs is called the dielectric strength of the dielectric material.

or

The dielectric strength is the maximum value of the electric field that a dielectric can tolerate or withstand without breakdown.

Continuity Equation and Relaxation Time

According to principle of charge conservation, the time rate of decrease of charge within a given volume must be equal to the net outward current flow through the closed surface of the volume.

The current I_{out} coming out of the closed surface

$$I_{\text{out}} = \oint \mathbf{J} \cdot d\mathbf{S} = \frac{-dQ_{\text{in}}}{dt} \quad (\text{i})$$

where Q_{in} is the total charge enclosed by the closed surface.

Using divergence theorem

$$\oint_S \mathbf{J} \cdot d\mathbf{S} = \int_v \nabla \cdot \mathbf{J} dv$$

But

$$\frac{-dQ_{\text{in}}}{dt} = -\frac{d}{dt} \int_v \rho_v dv = - \int_v \frac{\partial \rho_v}{\partial t} dv$$

Program Name: B.Tech.-CSE**Semester : I****Course Name : Physics for Computer Engineers****Time : 50 Minutes****Course Code : PHYS1036****Max Marks: 20****All questions are compulsory****Marks****CO**

Q 1	A three-level laser emits a light of wavelength of $5500 \text{ \AA}$. What will be the ratio of population of upper level E_2 to the lower energy level E_1 at the room temperature? (Given $h = 6.63 \times 10^{-34} \text{ Js}$, $k = 1.38 \times 10^{-23} \text{ J/K}$, $c = 3 \times 10^8 \text{ m/s}$)	4	CO1
Q 2	What is holography? Explain the re-construction of a hologram with suitable diagram.	4	CO1
Q 3	Explain single mode, multimode and graded index fibers?	4	CO1
Q 4	Define divergence of a vector field. Write the statement of Gauss-Divergence theorem and briefly explain its physical meaning.	4	CO2
Q 5	Find the gradient of the following scalar field: $V = e^{-z} \sin 2x \cosh v$	4	CO2

All questions are compulsory**Marks****CO**

Q 1	Explain the characteristics of a laser light.	4	CO1
Q 2	Mention any four differences between photography and holography.	4	CO1
Q 3	A signal of power 5 mW exists just at the entrance of 0.1 km long fibre. Calculate the attenuation coefficient of the fibre if the power at the end of the fiber is 1 mW.	4	CO1
Q 4	What is coulomb's law? Define electric field intensity and write expressions for the electric field for different charge distributions.	4	CO2
Q 5	Find the constant a , if the vector $\vec{A} = (x + 3y)\hat{i} + (2y + 3z)\hat{j} + (x + az)\hat{k}$ is a solenoidal vector.	4	CO2

Equation (i) now becomes

$$\int_v \nabla \cdot \mathbf{J} \, dv = - \int_v \frac{\partial \rho_v}{\partial t} \, dv$$

$$\text{or} \quad \nabla \cdot \mathbf{J} = - \frac{\partial \rho_v}{\partial t} \quad (\text{ii})$$

This is called the continuity of current equation.

Here, $\mathbf{J}$ is the current density and ρ is the charge density. This equation simply means that if charge is leaving a certain region (through current flow), the charge density inside that region decreases, and vice versa. In short, it ensures that charge is always conserved.

Effect of introducing charge at some interior point of a conductor/dielectric

According to Ohm's law $\mathbf{J} = \sigma \mathbf{E}$

According to Gauss's law $\nabla \cdot \mathbf{E} = \frac{\rho_v}{\epsilon}$

Equation (ii) now becomes

$$\nabla \cdot \sigma \mathbf{E} = \frac{\sigma \rho_v}{\epsilon} = -\frac{\partial \rho_v}{\partial t}$$

or
$$\frac{\partial \rho_v}{\partial t} + \frac{\sigma}{\epsilon} \rho_v = 0$$

This is homogeneous linear ordinary differential equation. By separating variables we get

$$\frac{\partial \rho_v}{\rho_v} = -\frac{\sigma}{\epsilon} \partial t$$

Integrating both sides

$$\ln \rho_v = -\frac{\sigma t}{\epsilon} + \ln \rho_{v0}$$

where $\ln \rho_{v0}$ is a constant of integration

$$\rho_v = \rho_{v0} e^{-t/T_r} \quad (\text{iii})$$

where

$$T_r = \frac{\epsilon}{\sigma}$$

ρ_{v0} is the initial charge density (i.e., ρ_v at $t = 0$)

Equation (iii) shows that as a result of introducing charge at some interior point of the material there is a decay of the volume charge density ρ_v .

The time constant T_r is known as the relaxation time or the relaxation time.

Relaxation time is the time in which a charge placed in the interior of a material to drop to $e^{-1} = 36.8 \%$ of its initial value.

For Copper $T_r = 1.53 \times 10^{-19}$ sec (short for good conductors)

For fused Quartz $T_r = 51.2$ days (large for good dielectrics)

Boundary Conditions

If the field exists in a region consisting of two different media, the conditions that the field must satisfy at the interface separating the media are called boundary conditions

These conditions are helpful in determining the field on one side of the boundary when the field on other side is known.

We will consider the boundary conditions at an interface separating

1. Dielectric (ϵ_{r1}) and Dielectric (ϵ_{r2})
2. Conductor and Dielectric
3. Conductor and free space

For determining boundary conditions we will use Maxwell's equations

$$\oint \mathbf{E} \cdot d\mathbf{l} = 0 \quad \text{and} \quad \oint \mathbf{D} \cdot d\mathbf{S} = Q_{\text{enc}}$$

Boundary Conditions (Between two different dielectrics)

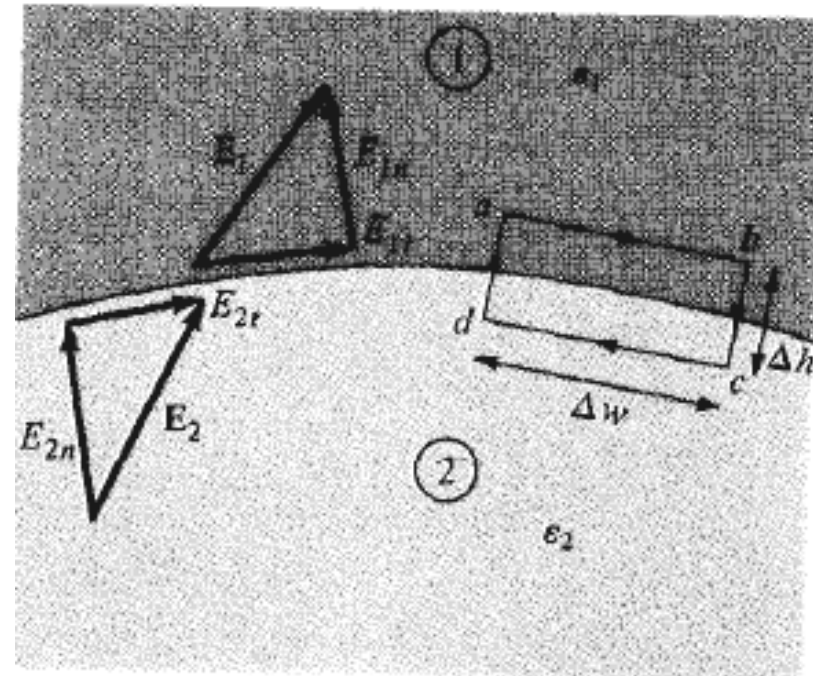
Consider the $\mathbf{E}$ field existing in a region consisting of two different dielectrics characterized by $\epsilon_1 = \epsilon_0 \epsilon_{r1}$ and $\epsilon_2 = \epsilon_0 \epsilon_{r2}$

$\mathbf{E}_1$ and $\mathbf{E}_2$ in the media 1 and 2 can be written as

$$\vec{E}_1 = \vec{E}_{1t} + \vec{E}_{1n} \quad \text{and} \quad \vec{E}_2 = \vec{E}_{2t} + \vec{E}_{2n}$$

But
$$\oint \mathbf{E} \cdot d\mathbf{l} = 0$$

Assuming that the path $abcda$ is very small with respect to the variation in $\mathbf{E}$



$$0 = E_{1t} \Delta w - E_{1n} \frac{\Delta h}{2} - E_{2n} \frac{\Delta h}{2} - E_{2t} \Delta w + E_{2n} \frac{\Delta h}{2} + E_{1n} \frac{\Delta h}{2}$$

As $\Delta h \rightarrow 0$

$$E_{1t} = E_{2t}$$

Thus the tangential components of $\mathbf{E}$ are the same on the two sides of the boundary. $\mathbf{E}$ is continuous across the boundary.

But $\mathbf{D} = \epsilon \mathbf{E} = \mathbf{D}_t + \mathbf{D}_n$

Thus

$$\frac{D_{1t}}{\epsilon_1} = E_{1t} = E_{2t} = \frac{D_{2t}}{\epsilon_2}$$

or
$$\frac{D_{1t}}{\epsilon_1} = \frac{D_{2t}}{\epsilon_2}$$

Here $\mathbf{D}_t$ undergoes some change across the surface and is said to be discontinuous across the surface.

Applying $\oint \mathbf{D} \cdot d\mathbf{S} = Q_{\text{enc}}$

Putting $\Delta h \rightarrow 0$ gives

$$\Delta Q = \rho_S \Delta S = D_{1n} \Delta S - D_{2n} \Delta S$$

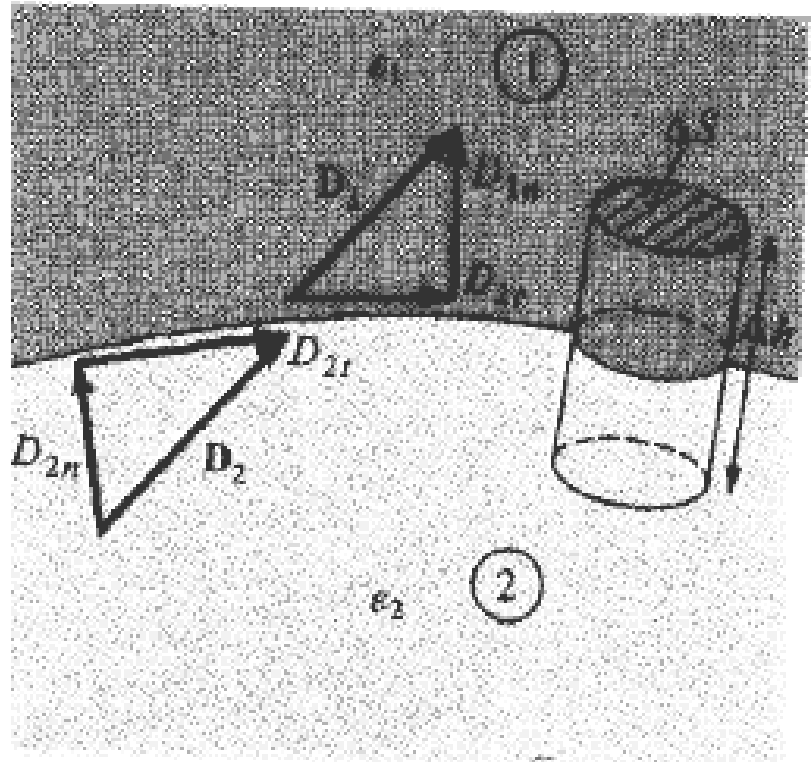
$$D_{1n} - D_{2n} = \rho_S$$

Where ρ_s is the free charge density placed deliberately at the boundary

If there is no charge on the boundary i.e. $\rho_s = 0$ then

$$D_{1n} = D_{2n}$$

Thus the normal components of $\mathbf{D}$ is continuous across the surface.



Analogy between Electric and Magnetic Field

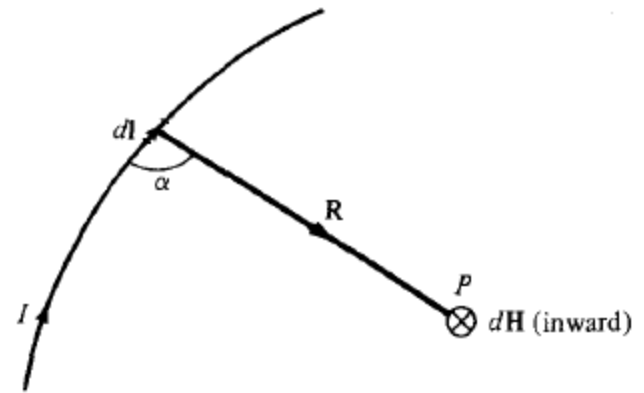
Term	Electric	Magnetic
Basic laws	$\mathbf{F} = \frac{Q_1 Q_2}{4\pi\epsilon_r^2} \mathbf{a}_r$	$d\mathbf{B} = \frac{\mu_0 I d\mathbf{l} \times \mathbf{a}_R}{4\pi R^2}$
	$\oint \mathbf{D} \cdot d\mathbf{S} = Q_{\text{enc}}$	$\oint \mathbf{H} \cdot d\mathbf{l} = I_{\text{enc}}$
Force law	$\mathbf{F} = QE$	$\mathbf{F} = Q\mathbf{u} \times \mathbf{B}$
Source element	dQ	$Q\mathbf{u} = I d\mathbf{l}$
Field intensity	$E = \frac{V}{\ell} \text{ (V/m)}$	$H = \frac{I}{\ell} \text{ (A/m)}$
Flux density	$\mathbf{D} = \frac{\Psi}{S} \text{ (C/m}^2\text{)}$	$\mathbf{B} = \frac{\Psi}{S} \text{ (Wb/m}^2\text{)}$
Relationship between fields	$\mathbf{D} = \epsilon\mathbf{E}$	$\mathbf{B} = \mu\mathbf{H}$
Potentials	$\mathbf{E} = -\nabla V$	$\mathbf{H} = -\nabla V_m \text{ (J = 0)}$
	$V = \int \frac{\rho_L d\mathbf{l}}{4\pi\epsilon r}$	$\mathbf{A} = \int \frac{\mu I d\mathbf{l}}{4\pi R}$
Flux	$\Psi = \int \mathbf{D} \cdot d\mathbf{S}$	$\Psi = \int \mathbf{B} \cdot d\mathbf{S}$
	$\Psi = Q = CV$	$\Psi = LI$
	$I = C \frac{dV}{dt}$	$V = L \frac{dI}{dt}$
Energy density	$w_E = \frac{1}{2} \mathbf{D} \cdot \mathbf{E}$	$w_m = \frac{1}{2} \mathbf{B} \cdot \mathbf{H}$

Biot-Savart's Law

It states that the magnetic field intensity $d\mathbf{H}$ produce at a point P by the differential current element $I d\mathbf{l}$ is proportional to the product $I d\mathbf{l}$ and the sine of angle α between the element and line joining P to the element and is inversely proportional to the square of distance R between P and the element.

$$dH \propto \frac{I d\mathbf{l} \sin \alpha}{R^2} \quad \text{or} \quad dH = \frac{I d\mathbf{l} \sin \alpha}{4\pi R^2}$$

$$d\mathbf{H} = \frac{I d\mathbf{l} \times \mathbf{a}_R}{4\pi R^2} = \frac{I d\mathbf{l} \times \mathbf{R}}{4\pi R^3}$$



The direction of $d\mathbf{H}$ can be determined by the right hand thumb rule with the right hand thumb pointing in the direction of the current, the right hand fingers encircling the wire in the direction of $d\mathbf{H}$

Biot-Savart's Law

If we define $\mathbf{K}$ as the surface charge density, $\mathbf{J}$ as volume current density then source elements are related as

$$I d\mathbf{l} \equiv \mathbf{K} dS \equiv \mathbf{J} dv$$

In terms of the distributed current sources, the Biot-Savart law becomes

$$\mathbf{H} = \int_L \frac{I d\mathbf{l} \times \mathbf{a}_R}{4\pi R^2} \quad (\text{line current})$$

$$\mathbf{H} = \int_S \frac{\mathbf{K} dS \times \mathbf{a}_R}{4\pi R^2} \quad (\text{surface current})$$

$$\mathbf{H} = \int_v \frac{\mathbf{J} dv \times \mathbf{a}_R}{4\pi R^2} \quad (\text{volume current})$$

Ampere's circuit Law

The line integral of the tangential component of $\mathbf{H}$ around a close path is the same as the net current I_{enc} enclosed by the path.

$$\oint \mathbf{H} \cdot d\mathbf{l} = I_{\text{enc}}$$

Using Stoke's law

$$I_{\text{enc}} = \oint_L \mathbf{H} \cdot d\mathbf{l} = \int_S (\nabla \times \mathbf{H}) \cdot d\mathbf{S}$$

But

$$I_{\text{enc}} = \int_S \mathbf{J} \cdot d\mathbf{S}$$

Comparing we get

$$\nabla \times \mathbf{H} = \mathbf{J}$$

This is third maxwell equation

Application of Ampere's Law : Infinite Line Current

Consider an infinitely long filamentary current I along the z -axis.

To calculate H at point P we will take a closed path passing through P known as Amperian path.

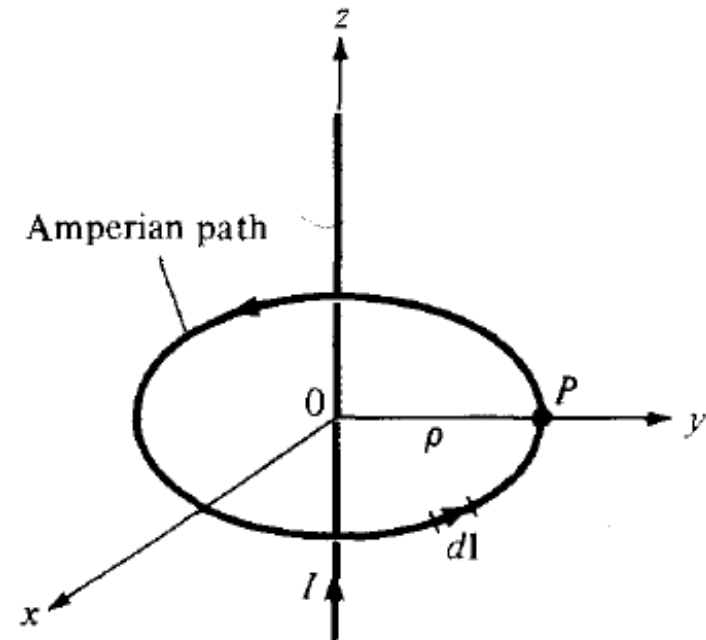
According to Ampere's Law

$$I = \int H_{\phi} \mathbf{a}_{\phi} \cdot \rho d\phi \mathbf{a}_{\phi}$$

$$= H_{\phi} \int \rho d\phi$$

$$= H_{\phi} \cdot 2\pi\rho$$

$$\mathbf{H} = \frac{I}{2\pi\rho} \mathbf{a}_{\phi}$$



Application of Ampere's law : Infinite Sheet Current

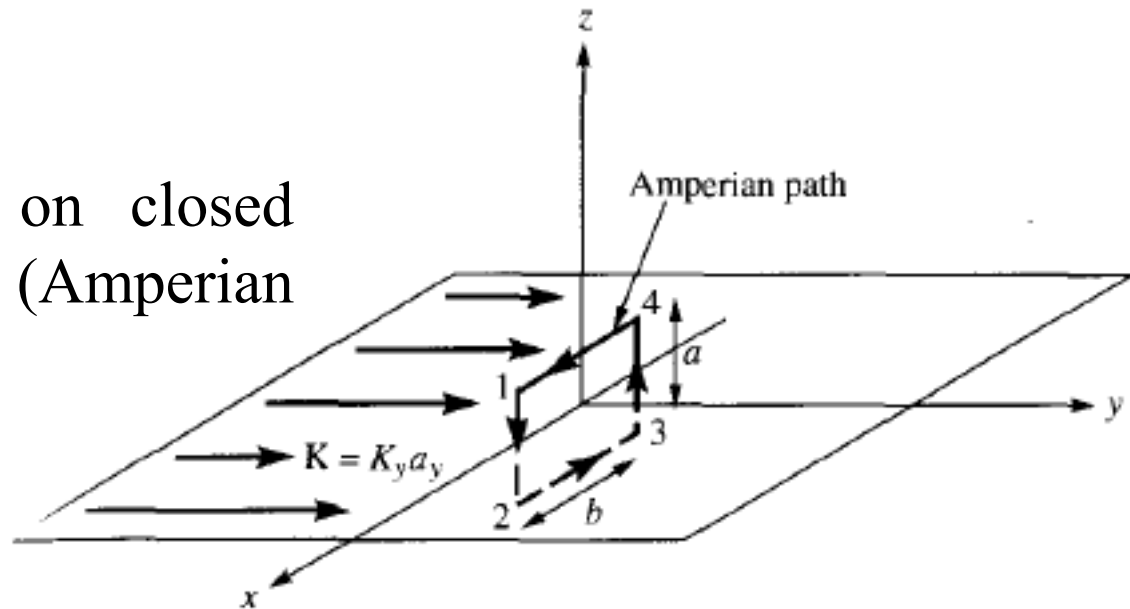
Consider an infinite current sheet in $z = 0$ plane.

If the sheet has a uniform current density then

$$\vec{K} = K_y \hat{a}_y$$

Applying Ampere's Law on closed rectangular path 1-2-3-4-1 (Amperian path) we get

$$\oint \mathbf{H} \cdot d\mathbf{l} = I_{enc} = K_y b \quad (i)$$



To solve integral we need to know how $\mathbf{H}$ is like

We assume the sheet comprising of filaments $d\mathbf{H}$ above and below the sheet due to pair of filamentary current.

The resultant $d\mathbf{H}$ has only an x-component.

Also $\mathbf{H}$ on one side of sheet is the negative of the other.

Due to infinite extent of the sheet, it can be regarded as consisting of such filamentary pairs so that the characteristic of $\mathbf{H}$ for a pair are the same for the infinite current sheets

$$\mathbf{H} = \begin{cases} H_0 \mathbf{a}_x & z > 0 \\ -H_0 \mathbf{a}_x & z < 0 \end{cases} \quad (\text{ii})$$

where H_0 is to be determined.

Evaluating the line integral of $\mathbf{H}$ along the closed path

$$\begin{aligned}\oint \mathbf{H} \cdot d\mathbf{l} &= \left(\int_1^2 + \int_2^3 + \int_3^4 + \int_4^1 \right) \mathbf{H} \cdot d\mathbf{l} \\ &= 0(-a) + (-H_o)(-b) + 0(a) + H_o(b) \\ &= 2H_o b \quad (\text{iii})\end{aligned}$$

Comparing (i) and (iii), we get

$$H_o = \frac{1}{2} K_y \quad (\text{iv})$$

Using (iv) in (ii), we get

$$\mathbf{H} = \begin{cases} \frac{1}{2} K_y \mathbf{a}_x, & z > 0 \\ -\frac{1}{2} K_y \mathbf{a}_x, & z < 0 \end{cases}$$

Generally, for an infinite sheet of current density $\mathbf{K}$

$$\mathbf{H} = \frac{1}{2} \mathbf{K} \times \mathbf{a}_n$$

where $\mathbf{a}_n$ is a unit normal vector directed from the current sheet to the point of interest.

Magnetic Flux Density

The magnetic flux density **B** is similar to the electric flux density **D**. Therefore, the magnetic flux density **B** is related to the magnetic field intensity **H**

$$\mathbf{B} = \mu_0 \mathbf{H}$$

where μ_0 is a constant and is known as the permeability of free space. Its unit is Henry/meter (H/m) and has the value

$$\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$$

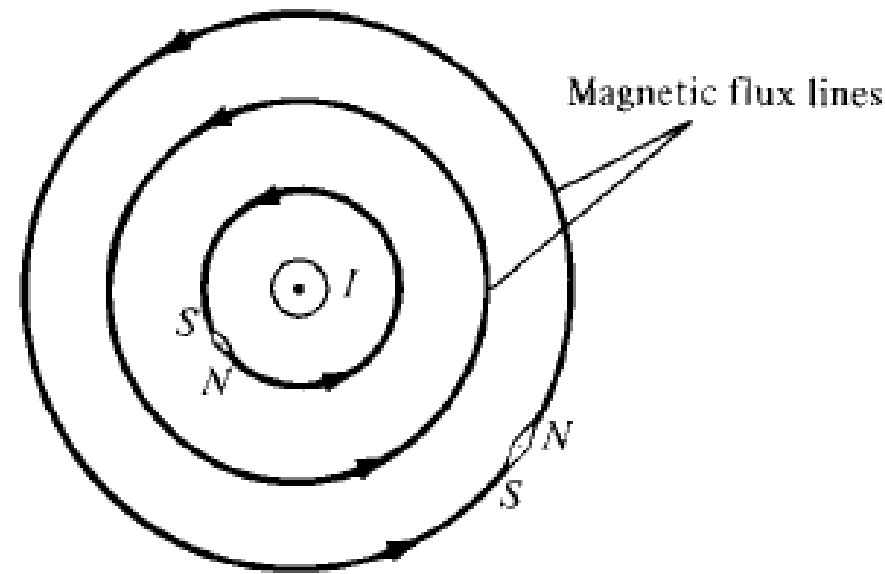
The magnetic flux through a surface **S** is given by

$$\Psi = \int_S \mathbf{B} \cdot d\mathbf{S}$$

where the magnetic flux ψ is in webers (Wb) and the magnetic flux density is in weber/ square meter or Teslas.

Magnetic flux lines due to a straight wire with current coming out of the page

Each magnetic flux line is closed with no beginning and no end and are also not crossing each other.

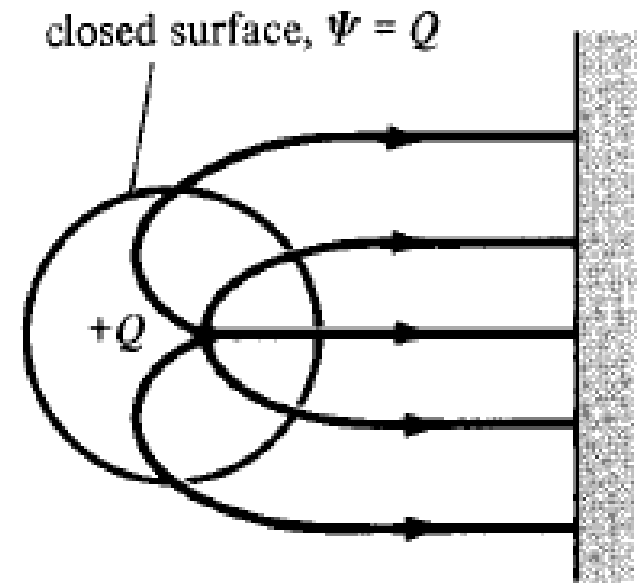


In an electrostatic field, the flux passing through a closed surface is the same as the charge enclosed.

$$\Psi = \oint \mathbf{D} \cdot d\mathbf{S} = Q$$

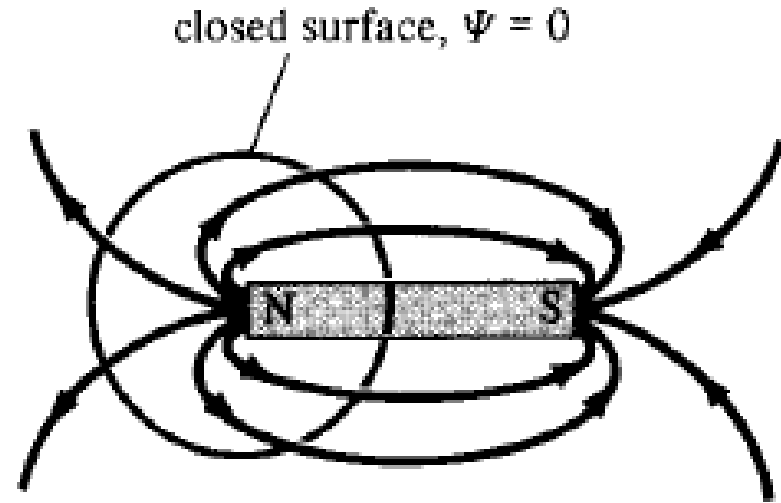
Thus it is possible to have an isolated electric charge.

Also the electric flux lines are not necessarily closed.



Magnetic flux lines are always closed upon themselves,.

So it is not possible to have an isolated magnetic pole (or magnetic charges)



An isolated magnetic charge does not exist.

Thus the total flux through a closed surface in a magnetic field must be zero.

$$\oint \mathbf{B} \cdot d\mathbf{S} = 0$$

This equation is known as the law of conservation of magnetic flux or Gauss's Law for Magnetostatic fields.

Magnetostatic field is not conservative but magnetic flux is conserved.

Applying Divergence theorem, we get

$$\oint_S \mathbf{B} \cdot d\mathbf{S} = \int_v \nabla \cdot \mathbf{B} dv = 0$$

or
$$\nabla \cdot \mathbf{B} = 0$$

This is Maxwell's fourth equation.

This equation suggests that magnetostatic fields have no source or sinks.

Also magnetic flux lines are always continuous.

Forces due to Magnetic Fields

There are at least three ways in which force due to magnetic fields can be experienced.

- a) Due to a moving charge particle in a magnetic field
- b) On a current element in an external magnetic field
- c) Between two current elements

Force on a charged particle

The force on a stationary or moving electric charge Q in an electric field is

$$\mathbf{F}_e = Q\mathbf{E}$$

The force on a charge moving with uniform velocity $\mathbf{u}$ in a magnetic field $\mathbf{B}$ is

$$\mathbf{F}_m = Q\mathbf{u} \times \mathbf{B}$$

Force on a Charged particle

The force on a moving charge Q in the presence of both electric field and magnetic field is given by

$$\mathbf{F} = \mathbf{F}_e + \mathbf{F}_m$$

$$\mathbf{F} = Q(\mathbf{E} + \mathbf{u} \times \mathbf{B})$$

This is known as Lorentz force equation.

If the mass of the charged particle moving in E and B fields is m , by Newton's law of motion

$$\mathbf{F} = m \frac{d\mathbf{u}}{dt} = Q (\mathbf{E} + \mathbf{u} \times \mathbf{B})$$

Magnetic Moment and Torque

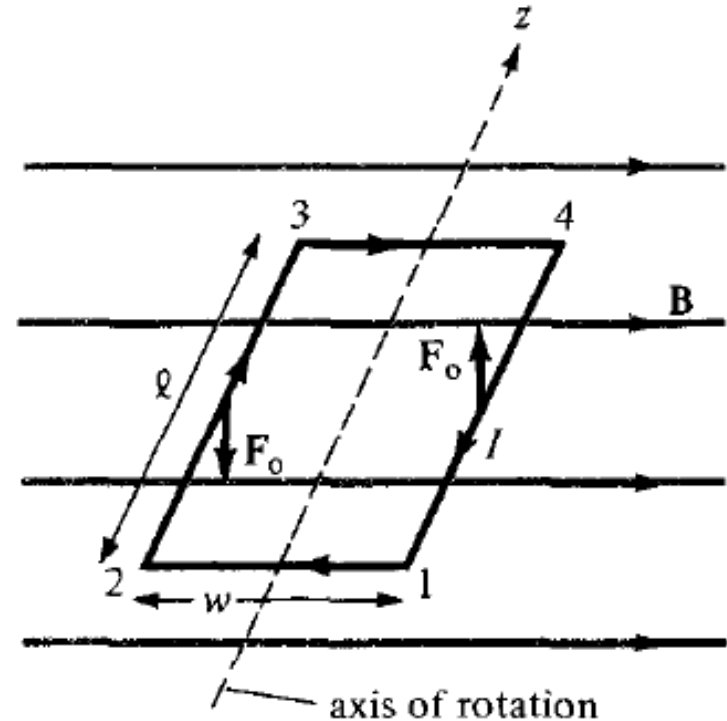
Consider a rectangular loop of length l and width w placed in a uniform magnetic field $\mathbf{B}$.

Here $d\mathbf{l}$ is parallel to $\mathbf{B}$ along sides 12 and 34 of the loop and hence no force is exerted on these sides

$$\mathbf{F} = I \int_2^3 d\mathbf{l} \times \mathbf{B} + I \int_4^1 d\mathbf{l} \times \mathbf{B}$$

$$= I \int_0^\ell dz \mathbf{a}_z \times \mathbf{B} + I \int_\ell^0 dz \mathbf{a}_z \times \mathbf{B}$$

$$\mathbf{F} = \mathbf{F}_0 - \mathbf{F}_0 = 0$$



Magnetic Moment and Torque

Where $|\mathbf{F}_o| = IB\ell$ because $\mathbf{B}$ is uniform.

Thus no force is exerted on the loop as a whole. However $\mathbf{F}_o$ and $-\mathbf{F}_o$ act at different points, creating a couple.

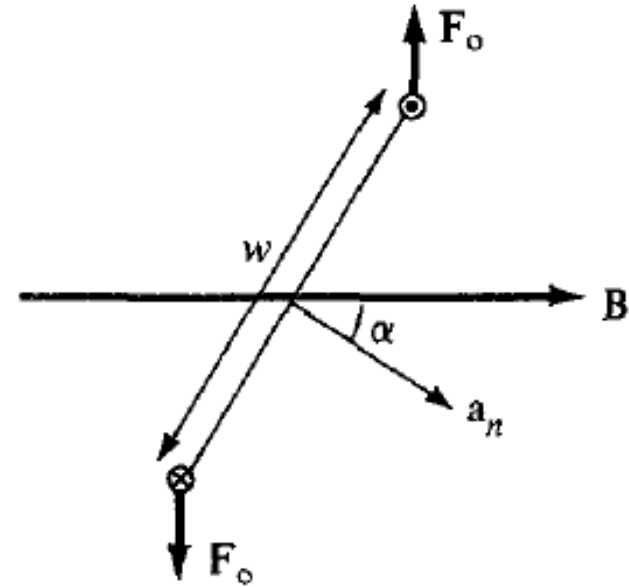
If normal to the plane of the loop makes an angle α with $\mathbf{B}$, the torque on the loop is

$$|\mathbf{T}| = |\mathbf{F}_o| w \sin \alpha$$

$$T = BI\ell w \sin \alpha$$

But $\ell w = S$, area of the loop

$$T = BIS \sin \alpha$$



Magnetic Moment and Torque

Now we define

$$\mathbf{m} = IS\mathbf{a}_n$$

as the dipole moment (in A.m²) of the loop.

The magnetic dipole moment is the product of current and area of the loop and its direction is normal to the loop.

$$\mathbf{T} = \mathbf{m} \times \mathbf{B}$$

We can apply this equation for loop of any arbitrary shape. The only limitation is that the magnetic field should be uniform.

Faraday's Law

According to Faraday a time varying magnetic field produces an induced voltage (called electromotive force or emf) in a closed circuit, which causes a flow of current.

The induced emf (V_{emf}) in any closed circuit is equal to the time rate of change of the magnetic flux linkage by the circuit. This is Faraday's Law and can be expressed as

$$V_{\text{emf}} = -N \frac{d\Psi}{dt}$$

where N is the number of turns in the circuit and ψ is the flux through each turn.

The negative sign shows that the induced voltage acts in such a way to oppose the flux producing in it. This is known as Lenz's Law.

Transformer and Motional EMF

For a circuit with a single turn ($N = 1$)

$$V_{\text{emf}} = -\frac{d\Psi}{dt}$$

In terms of $\mathbf{E}$ and $\mathbf{B}$ this can be written as

$$V_{\text{emf}} = \oint_L \mathbf{E} \cdot d\mathbf{l} = -\frac{d}{dt} \int_S \mathbf{B} \cdot d\mathbf{S} \quad (\text{i})$$

where ψ has been replaced by $\int_S \mathbf{B} \cdot d\mathbf{S}$ and S is the surface area of the circuit bounded by a closed path L ..

The equation says that in time-varying situation, both electric and magnetic fields are present and are interrelated.

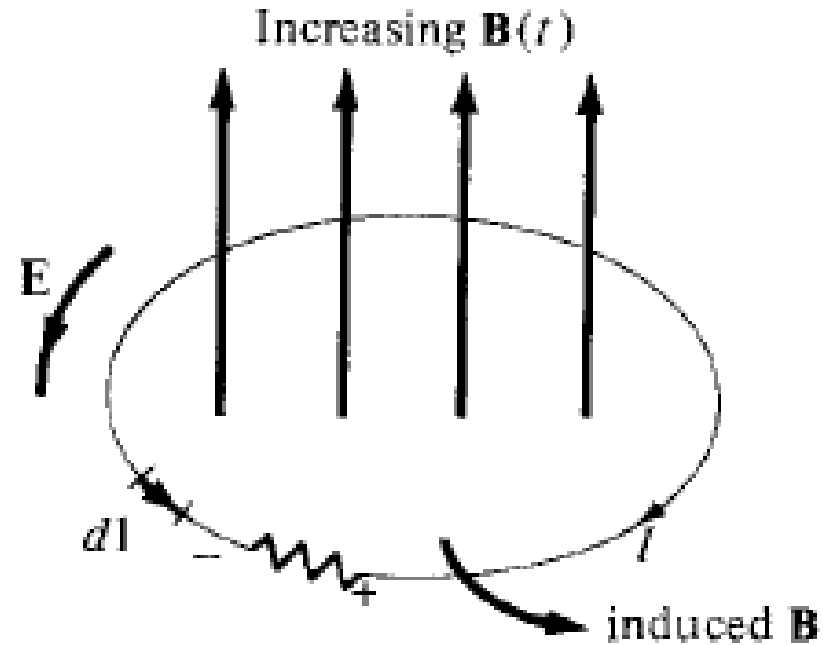
The variation of flux with time may be caused in three ways.

1. By having a stationary loop in a time-varying $\mathbf{B}$ field.
2. By having a time-varying loop area in a static $\mathbf{B}$ field.
3. By having a time-varying loop area in a time-varying $\mathbf{B}$ field.

Stationary loop in a time-varying $\mathbf{B}$ field (Transformer emf)

Consider a stationary conducting loop in a time-varying magnetic $\mathbf{B}$ field. The equation (i) becomes

$$V_{\text{emf}} = \oint_L \mathbf{E} \cdot d\mathbf{l} = - \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{S}$$



This emf induced by the time-varying current in a stationary loop is often referred to as transformer emf in power analysis since it is due to the transformer action.

By applying Stokes's theorem to the middle term, we get

$$\int_S (\nabla \times \mathbf{E}) \cdot d\mathbf{S} = - \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{S}$$

Thus

$$\nabla \times \mathbf{E} = - \frac{\partial \mathbf{B}}{\partial t}$$

This is one of the Maxwell's equations for time-varying fields.

It shows that the time-varying field is not conservative.

$$\nabla \times \mathbf{E} \neq 0$$

2. Moving loop in static $\mathbf{B}$ field (Motional emf)

When a conducting loop is moving in a static $\mathbf{B}$ field, an emf is introduced in the loop.

The force on a charge moving with uniform velocity $\mathbf{u}$ in a magnetic field $\mathbf{B}$ is

$$\mathbf{F}_m = Q\mathbf{u} \times \mathbf{B}$$

The motional electric field $\mathbf{E}_m$ is defined as

$$\mathbf{E}_m = \frac{\mathbf{F}_m}{Q} = \mathbf{u} \times \mathbf{B}$$

Consider a conducting loop moving with uniform velocity $\mathbf{u}$, the emf induced in the loop is

$$V_{\text{emf}} = \oint_L \mathbf{E}_m \cdot d\mathbf{l} = \oint_L (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l} \quad (\text{i})$$

This kind of emf is called the motional emf or flux-cutting emf. Because it is due to the motional action. eg., Motors, generators

By applying Stokes's theorem to equation (i), we get

$$\int_S (\nabla \times \mathbf{E}_m) \cdot d\mathbf{S} = \int_S \nabla \times (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{S}$$

$$\nabla \times \mathbf{E}_m = \nabla \times (\mathbf{u} \times \mathbf{B})$$

3. Moving loop in time-varying field

Consider a moving conducting loop in a time-varying magnetic field

Then both transformer emf and motional emf are present.

Thus the total emf will be the sum of transformer emf and motional emf

$$V_{\text{emf}} = \oint_L \mathbf{E} \cdot d\mathbf{l} = - \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{S} + \oint_L (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l}$$

also

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} + \nabla \times (\mathbf{u} \times \mathbf{B})$$

Displacement Current

For static EM fields

$$\nabla \times \mathbf{H} = \mathbf{J} \quad (\text{i})$$

But the divergence of the curl of a vector field is zero. So

$$\nabla \cdot (\nabla \times \mathbf{H}) = 0 = \nabla \cdot \mathbf{J} \quad (\text{ii})$$

But the continuity of current requires

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho_v}{\partial t} \neq 0 \quad (\text{iii})$$

Equation (ii) and (iii) are incompatible for time-varying conditions

So we need to modify equation (i) to agree with (iii)

Add a term to equation (i) so that it becomes

$$\nabla \times \mathbf{H} = \mathbf{J} + \mathbf{J}_d \quad (\text{iv})$$

where $\mathbf{J}_d$ is to be defined and determined.

Again the divergence of the curl of a vector field is zero. So

$$\nabla \cdot (\nabla \times \mathbf{H}) = 0 = \nabla \cdot \mathbf{J} + \nabla \cdot \mathbf{J}_d \quad (\text{v})$$

In order for equation (v) to agree with (iii)

$$\nabla \cdot \mathbf{J}_d = -\nabla \cdot \mathbf{J} = \frac{\partial \rho_v}{\partial t} = \frac{\partial}{\partial t} (\nabla \cdot \mathbf{D}) = \nabla \cdot \frac{\partial \mathbf{D}}{\partial t}$$

$$\text{or} \quad \mathbf{J}_d = \frac{\partial \mathbf{D}}{\partial t} \quad (\text{vi})$$

Putting (vi) in (iv), we get

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

This is Maxwell's equation (based on Ampere Circuital Law) for a time-varying field. The term $\mathbf{J}_d = \partial \mathbf{D} / \partial t$ is known as displacement current density and $\mathbf{J}$ is the conduction current density $\mathbf{J} = \sigma \mathbf{E}$.

Maxwell's Equations in Final Form

Differential Form	Integral Form	Remarks
$\nabla \cdot \mathbf{D} = \rho_v$	$\oint_S \mathbf{D} \cdot d\mathbf{S} = \int_v \rho_v dv$	Gauss's law
$\nabla \cdot \mathbf{B} = 0$	$\oint_S \mathbf{B} \cdot d\mathbf{S} = 0$	Nonexistence of isolated magnetic charge*
$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$	$\oint_L \mathbf{E} \cdot d\mathbf{l} = -\frac{\partial}{\partial t} \int_S \mathbf{B} \cdot d\mathbf{S}$	Faraday's law
$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$	$\oint_L \mathbf{H} \cdot d\mathbf{l} = \int_S \left(\mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} \right) \cdot d\mathbf{S}$	Ampere's circuit law

Electric Flux Density

For an infinite sheet the electric flux density **D** is given by

$$\mathbf{D} = \frac{\rho_S}{2} \mathbf{a}_n$$

For a volume charge distribution the electric flux density **D** is given by

$$\mathbf{D} = \int \frac{\rho_v dv}{4\pi R^2} \mathbf{a}_R$$

In both the above equations **D** is a function of charge and position only (independent of medium) **slide no. 12**